IKT215 Assignment 2: Parametric and Non-parametric Classification

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Course: IKT215

Disclaimer on Generative AI

In accordance with UiA's guidelines, I declare that I used Generative AI (Different across the research process) to assist in structuring the LaTeX report and generating some of the boilerplate plotting code. The QDA and kNN implementations, the feature selection reasoning, and the analysis of results are my own work.

Introduction

This assignment is about implementing two classifiers from scratch and comparing them on the Wine Quality dataset. The first is a parametric plug-in classifier using Quadratic Discriminant Analysis (QDA), and the second is a non-parametric k-Nearest Neighbors (kNN) classifier. Both classify wines into *good* (quality ≥ 5) and *bad* (quality < 5).

I used the combined red and white wine datasets (6497 samples, 1599 red + 4898 white), split 80/20 into training (5197) and testing (1300). The classes are very imbalanced — only about 3.8% of the samples are "bad", which has a big effect on both classifiers.

Feature Selection (shared by both tasks)

I picked 4 features: **alcohol**, **volatile acidity**, **sulphates**, and **citric acid**. Two reasons:

1. **Correlation with quality:** Alcohol has the strongest correlation ($|r| = 0.44$), followed by volatile acidity ($|r| = 0.27$). Density also correlates well ($|r| = 0.31$), but it is highly correlated with alcohol ($r = -0.69$), so including both would be redundant.

2. **Low inter-feature correlation:** The selected features have low pairwise correlations (highest: volatile acidity vs citric acid at $|r| = 0.38$), so they each bring relatively independent information.

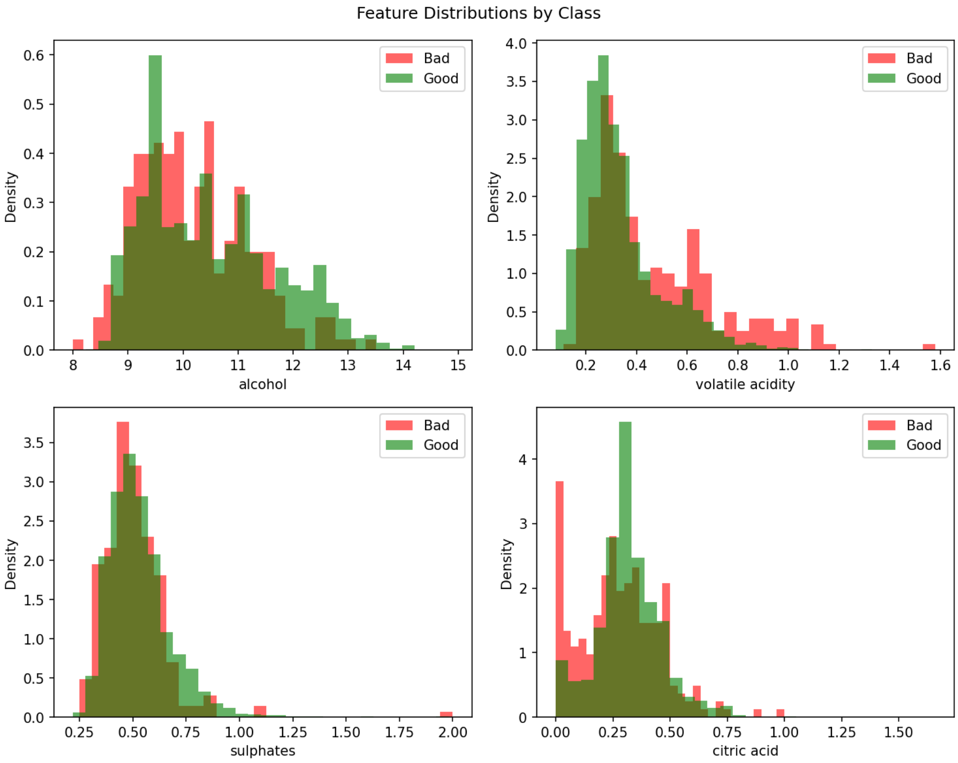

Figure 1: Feature distributions by class. Volatile acidity shows the clearest separation.

Task 1: Quadratic Discriminant Analysis

Parameter Estimation (MLE)

QDA assumes each class follows a multivariate Gaussian with its own mean and covariance. I estimated these using Maximum Likelihood Estimation:

Priors: $P(\text{bad}) = 195/5197 = 0.0375, \quad P(\text{good}) = 5002/5197 = 0.9625$

MLE Mean Vectors (features in order: alcohol, volatile acidity, sulphates, citric acid):

$$\hat{\boldsymbol{\mu}}_{\text{bad}} = (10.15,\ 0.459,\ 0.509,\ 0.273)^T, \quad \hat{\boldsymbol{\mu}}_{\text{good}} = (10.50,\ 0.336,\ 0.534,\ 0.320)^T$$

MLE Covariance Matrices (using $\frac{1}{N}$ normalization):

$$\hat{\Sigma}_{\text{bad}} = \begin{pmatrix} 0.990 & 0.008 & -0.013 & -0.018 \\ 0.008 & 0.059 & 0.004 & -0.019 \\ -0.013 & 0.004 & 0.027 & 0.007 \\ -0.018 & -0.019 & 0.007 & 0.032 \end{pmatrix}, \quad \hat{\Sigma}_{\text{good}} = \begin{pmatrix} 1.433 & -0.008 & -0.004 & -0.004 \\ -0.008 & 0.025 & 0.006 & -0.008 \\ -0.004 & 0.006 & 0.023 & 0.001 \\ -0.004 & -0.008 & 0.001 & 0.020 \end{pmatrix}$$

The covariance matrices are clearly different (e.g., alcohol variance is 0.99 for bad vs 1.43 for good), which is why QDA is more appropriate here than LDA.

Classifier Definition

The QDA discriminant function for class c is:

$$g_c(\mathbf{x}) = -\frac{1}{2}\ln|\hat{\Sigma}_c| - \frac{1}{2}(\mathbf{x} - \hat{\boldsymbol{\mu}}_c)^T \hat{\Sigma}_c^{-1}(\mathbf{x} - \hat{\boldsymbol{\mu}}_c) + \ln P(c) \tag{1}$$

The classifier assigns the class with the highest discriminant score.

Performance Evaluation

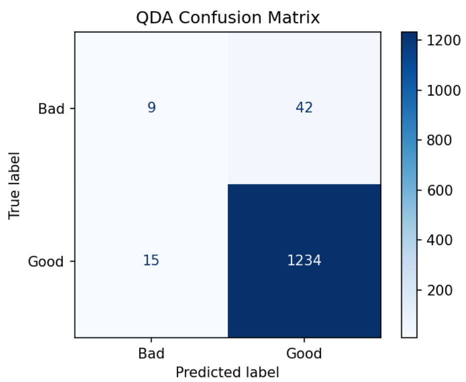

Figure 2: QDA confusion matrix. It catches 9 out of 51 bad wines.

QDA achieved **95.62%** overall accuracy, but this is misleading. Looking at per-class metrics: the classifier only catches 18% of the bad wines (recall = 0.18), and when it does predict "bad", it is only right 38% of the time (precision = 0.38). It is basically good at the majority class but struggles with the minority. This is a direct consequence of the heavy class imbalance and the strong prior toward "good".

Task 2: k-Nearest Neighbors Classifier

Standardization

Since kNN is distance-based, I standardized all features to zero mean and unit variance using the training set statistics. Without this, alcohol (range ∼8–15) would dominate over sulphates (range ∼0.2–2.0).

Distance Metric and k Selection

I used **Euclidean distance**: $d(\mathbf{x}, \mathbf{x}') = \sqrt{\sum_i (x_i - x_i')^2}$. This is the standard choice and works well since all features are continuous and on the same scale after standardization.

For k, I tested values from 1 to 31 (Figure 3). I went with $k = 5$ because accuracy stabilizes around $k = 5$–7. Smaller values like $k = 1$ are prone to overfitting, and going much higher makes the classifier default to the majority class even more.

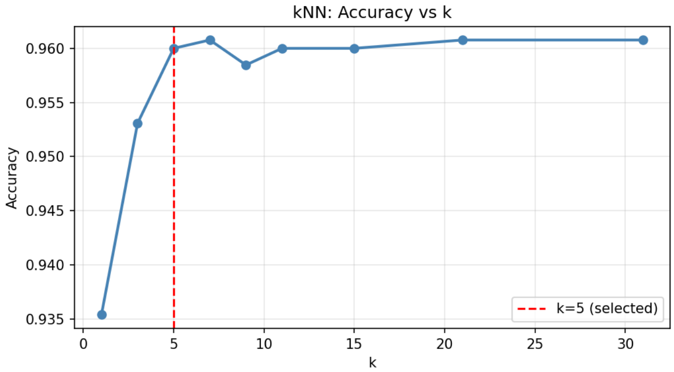

Figure 3: kNN accuracy for different k values. Stabilizes around $k = 5$.

Performance Evaluation

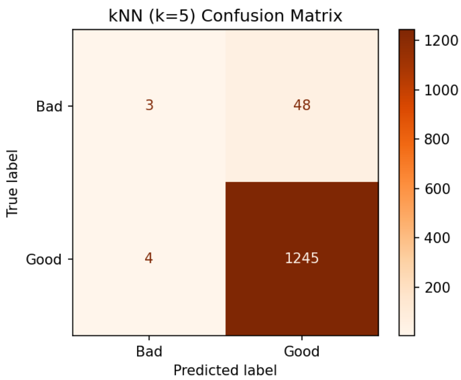

Figure 4: kNN ($k = 5$) confusion matrix. Only catches 3 out of 51 bad wines.

kNN achieved **96.00%** accuracy — slightly higher than QDA in raw numbers, but actually worse at detecting bad wines (recall = 0.06 vs QDA's 0.18).

Comparison: QDA vs kNN

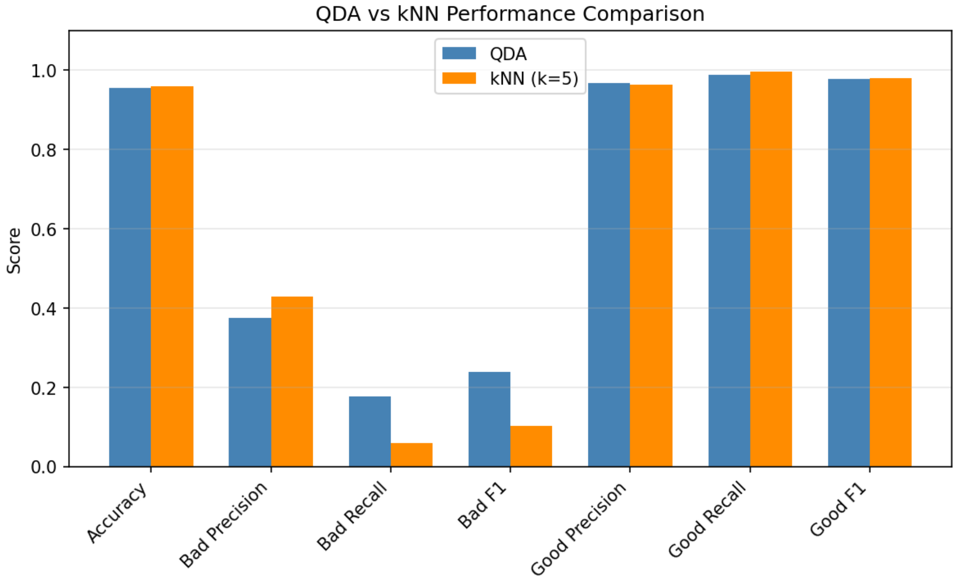

Figure 5: Side-by-side performance comparison. QDA has better minority class recall.

Table 1: Summary comparison of QDA and kNN.

Metric	QDA	kNN ($k = 5$)
Overall Accuracy	0.9562	0.9600
Bad Precision / Recall / F1	0.38 / 0.18 / 0.24	0.43 / 0.06 / 0.10
Good Precision / Recall / F1	0.97 / 0.99 / 0.98	0.96 / 1.00 / 0.98

Key observations:

- **QDA is better at detecting bad wines.** Higher recall (0.18 vs 0.06) and F1 (0.24 vs 0.10) for the minority class. This is likely because QDA explicitly models the class distributions and uses the prior, so it captures the structure of the "bad" class better.
- **kNN defaults to the majority class.** Since 96% of training neighbors are "good", most query points get outvoted.
- **The class imbalance is the elephant in the room.** Neither classifier handles it well. Techniques like oversampling or threshold adjustment would help, but were outside the scope.

Conclusion

I implemented QDA and kNN from scratch and tested them on the Wine Quality dataset. Both achieve ∼96% accuracy, but that number is misleading given the 96/4 class imbalance. QDA performs better on the minority class because it explicitly models the class-conditional distributions rather than relying on local neighborhoods. The main takeaway is that accuracy alone is not a good metric when classes are imbalanced — precision, recall, and F1 per class tell a much more honest story.